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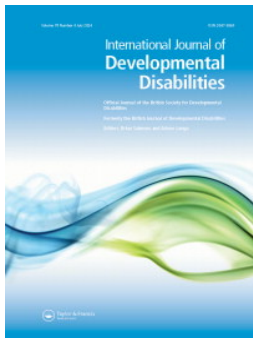


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The role of physical activity in social and behavioral skills of children with autism spectrum disorder: a case-controlled study

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ABSTRACT

Objectives: Despite the numerous benefits of regular participation in physical activities for health and social engagement, children with autism spectrum disorder (ASD) who experience difficulties in social and behavioral skills are less likely to participate in physical activities compared to their peers. There is a need for more research on the impact of regular participation in physical activities on social and behavioral skills in children diagnosed with ASD. This study aimed to compare the social and behavioral skills of children diagnosed with ASD who participate in physical activity with those who do not.

Methods: The participants are parents of children diagnosed with ASD ($n = 60$). The study group ($n = 30$; 12.33 ± 1.58 years) attended a rehabilitation center and a sports club, and the control group ($n = 30$; 12.57 ± 1.52 years) attended a rehabilitation center constituted two study groups. The Autism Behavior Checklist was used to assess the behavioral skills of children diagnosed with ASD aged 10–15, and the Social Skills Assessment Scale was used to evaluate their social skill levels based on their participation in physical activity.

Results: According to the analysis, statistically significant differences were found in the social skill scale and behavioral skills of children who participated in physical activity compared to those who did not. ($p < .001$). Additionally, the prediction of social skills' behavior was examined by regression analysis. Accordingly, social skills predicted behaviors in the group participating in physical activity; it was found that it did not predict the group that did not participate in physical activity.

Conclusions: There was a difference between the behavioral and social skills of children diagnosed with ASD between the ages of 10–15 who do and do not participate in physical activity. This was an indication that physical activity was a positive aspect. It is recommended that future studies examine the effect of physical activity on children with autism more thoroughly.

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Introduction

Autism Spectrum Disorder (ASD) is recognized as a neurodevelopmental disorder characterized by problems in behavior, communication, and social relationships, beginning in the early stages of life and persisting throughout the lifespan (American Psychiatric Association 2013; Bremer, Crozier, and Lloyd 2016). The symptoms that begin in early childhood include difficulties in social interaction and communication and limited and repetitive behavior patterns (APA 2013; National Autism Center 2015). In today's world, the prevalence of ASD is increasing every year (Zeidan et al. 2022). Although it was historically considered purely as a psychological or neurological disorder, there is growing evidence that indicates ASD not only affects individuals psychologically but also has physical

implications (Garcia-Pastor et al. 2019; Staples and Reid 2010). Considering the descriptive symptoms of ASD, it is noted that these children encounter various difficulties throughout their development compared to their neurotypical children, (Genovese and Butler 2023). In addition, it was determined that there is a high relationship between manual skills and social skills. It was stated that children with ASD who have low manual skills also have low social skills (Hirata et al. 2014). These are defined as difficulty in establishing appropriate communication with peers, providing relative flexibility in daily activities, achieving behavioral and emotional maturity, developing age-appropriate peer relationships, and experiencing challenges in perceiving, interpreting, and navigating the social world successfully in their environment (Seltzer et al. 2003; Storch et al. 2012). These difficulties have the

potential to bring about long-term physiological, mental, and psychological health issues (Chu et al. 2020). Various interventions are recommended in the literature to cope with these difficulties (Adams et al. 2012; Bumin et al. 2015; Bohlander, Orlich, and Varley 2012; Draudvilienė et al. 2020; Flynn and Healy 2012; Hynes and Block 2022; Minshawi et al. 2016; Mandecka and Regulska-Ilow 2022; Rotheram-Fuller and MacMullen 2011; Sansosti 2010). Foremost among them are behavior management therapies incorporating Applied Behavior Analysis (ABA) (Flynn and Healy 2012) and Positive Behavioral Support (PBS) (Sansosti 2010), cognitive-behavioral therapy (Rotheram-Fuller and MacMullen 2011) focusing on the connection between thoughts and feelings, social skills training (Bohlander, Orlich, and Varley 2012), medication therapy (Minshawi et al. 2016), nutritional therapy (Mandecka and Regulska-Ilow 2022), and practices such as speech and language therapy (Adams et al. 2012), occupational therapy (Bumin et al. 2015), and physiotherapy (Draudvilienė et al. 2020; Hynes and Block 2022). In addition to these approaches, it was shown that physical activities improve social, behavioral, and academic skills in children and adolescents with ASD (Arnell, Jerlinder, and Lundqvist 2020; Healy et al. 2018; Sansi et al. 2021; Tiner, Cunningham, and Pittman 2021).

Physical activity is energy expenditure that involves unsystematic and unstructured movements at places such as homes, schools, parks, streets, and shopping malls (Bates et al. 2020; Hendricks et al. 2014). The benefits of physical activity for children and adolescents include improved cardiometabolic health, bone density, motor skills, emotional regulation, and psychological well-being (Janssen and LeBlanc 2010; Saunders et al. 2016). Although there are restrictions on physical activities, especially in children with ASD (Gürkan and Koçak 2023), physical activity can provide a natural environment for individuals to build relationships, increase opportunities for interaction, and offer the chance to participate in collaboration through play or teamwork (Zhao et al. 2021). Furthermore, regular participation in physical activity was associated with improved psychological well-being (LaVigne et al. 2016), self-concept, concentration, memory, and academic performance (Batista et al. 2022), as well as reduced anxiety and depression (Murphy, Sweeney, and McGrane 2020) in neurotypical children. For neurotypical children (Rhodes, Cox, and Sayar 2022) or children with intellectual disabilities (Borland et al. 2022), these studies often point to the social and behavioral benefits of physical activity.

Holloway and Long (2019) emphasize that participation in physical activities provides skill development and increases independence in children with ASD in treatment methods applied for physical activity in children with ASD. Tomporowski (2003) explained the short-term benefits of physical activities and behavioral techniques as therapies for adolescents with difficulty focusing and controlling impulsive actions. A systematic review reported that children with ASD participated intensively in physical activities such as swimming, cycling, neuromuscular training, yoga, sports, exercise, and games. These activities improved social skills such as language, eye contact, and speech. It emphasizes the need for more comprehensive research on the relationship between physical activity and social skills, including relationship building, social and self-care skills, and language skills, especially in children with ASD (Hynes and Block 2022). There was a call for more research on its effects on behaviors such as problem-solving, planning, and interaction (Bremer, Crozier, and Lloyd 2016; Holloway and Long 2019; Yarımkaaya and Esentürk 2020). Hynes and Block (2022) highlighted the gaps in understanding the relationship between physical activity and behaviors, communication, and social skills. They suggested that the relationship between physical activity and ASD characteristics should be explained in detail (Hynes and Block 2022). It was also noted that parents and clinicians recommend physical activity as an intervention alternative to improve social, behavioral, and academic skills in children and young adults with ASD (Hynes and Block 2022). Therefore, it was stated in the study that children diagnosed with ASD should be directed to physical activities.

In the literature on physical activity in children with ASD, it was stated that opportunities to participate in physical activities were limited (Anderson et al. 2004; Memari et al. 2014), and the most important reason was social interaction and communication difficulties during physical activity. (Pan 2008, 2009; Pan et al. 2011; Yarımkaaya and Esentürk 2022). Therefore, children with ASD tend to engage more in passive activities (such as solitary play, parallel play, and orientation) in their free time (Wolfberg et al. 2015). As a cause of these disadvantages, it was stated that there were limited studies on the effects of physical activity in children with ASD and that more studies should be conducted on the subject in the existing studies. This study aimed to compare the social and behavioral skills of children diagnosed with ASD who participate in physical activity with those who do not.

Material and method

Model of the research

This research is a quantitative study designed as a case-control study. Case-control studies are retrospective analyses conducted to determine what is desired to be investigated in two groups exposed to a factor or disease (Özdemir 2022). Our study compares the current social and behavioral skills of children with ASD who participated and did not participate in physical activities for a certain period in the past.

Participants

This study was conducted with the parents of ASD children who applied to various sports centers in Ankara and the pediatric unit of the Department of Occupational Therapy, Faculty of Health Sciences, Hacettepe University. The inclusion criteria for the group of children participating in physical activity were (1) having a diagnosis of ASD, (2) being between the ages of 10–15 years, and (3) participating in physical activity regularly. Exclusion criteria were (1) having a concomitant neurological, psychiatric, or orthopedic diagnosis and (2) having a physical activity duration of less than one month. The inclusion criteria for the control group were as follows: (1) having a diagnosis of ASD, (2) being between the ages of 10–15 years, and the exclusion criteria were as follows: (1) having a concomitant neurological, psychiatric, or orthopedic diagnosis, (2) being participated in regular physical activity. The number of children with ASD and their parents who applied to participate in the study at the sports center was 37. Five of them were excluded from the study because they did not participate in physical activity regularly. The parents of two of them did not fill the scales. In general, the children who benefited from physical activities in this sports school participated in training hours (1.5 h*2) twice a week. They were not taking part in any competitions. The number of children with ASD and their parents who applied to take part in the control group was 33. The parents of 3 children with ASD were not included in the study because they had comorbid neurological diseases. In total, the study was conducted with 60 parents of children with ASD (parents of children with ASD who did sports $n=30$; parents of children with autism who did not do sports $n=30$).

Study design

This study, conducted as case-control research, received ethical approval from the the Hacettepe University

Non-Interventional Clinical Research Ethics Board with the decision number 2022/12-04 on 05.07.2022. As a result of the power analysis performed with the GPower 3.1.9.7 program, for the determination of the difference between the groups in terms of the Autism Behaviour Checklist -ABC measurement, type 1 error: 0.05, effect size = 3.266 and power level = 1.000 with 30 participants in the groups; for the determination of the difference between the groups in terms of the Social Skills Assessment Scale measurement, type 1 error: 0.05, effect size = 3.289 and power level = 1.000 with 30 participant in the groups (Faul et al. 2009). Participants ($n=60$) (parents) were provided information about the research, and the informed consent form was read and signed. Data collection tools included a Socio-demographic information form, an Autism Behavior Checklist, and a Social Skills Assessment Scale.

Data collection tools

Sociodemographic information form

A sociodemographic information form created by the researcher was used to gather information about the participants in the study. This survey form included questions about the participant child's age, gender, participation in physical activity, the types of physical activity, and the duration of their involvement. Additionally, the form included questions about the participant's parent's age, gender, degree of relationship with the children, and monthly household income.

Autism behavior checklist – ABC

This scale, developed by Krug, Arick, and Almond (1993), was used to gather information about behaviors in individuals with ASD. The scale's Turkish validity and reliability study was conducted by Irmak et al. (2007). The scale consists of 57 items and five sub-dimensions, including sensory (9 items), relational (12 items), body and object use (12 items), language skills (13 items), and social and self-care skills (11 items). The scale is administered to parents and teachers of children diagnosed with ASD. Possible scores on the scale can range from 0 to 159. The internal consistency level has a Cronbach Alpha coefficient of .88. The test-retest correlation coefficient is 0.99 for the teacher form, 0.76 for the parent form, and 0.55 for teachers and parents. A higher score on this test means higher autism in terms of behavioral skills (Özdemir et al. 2014).

Social skills assessment scale

This scale, developed by Ataş, Efeçinar, and Tatar (2016), was used to gather information about social skills in children with ASD. Parents, teachers, or another close relative can fill out the scale. Comprising 76 items, the scale consists of seven sub-dimensions, namely 'Initiating and Maintaining Relationship Skills,' 'Assertiveness Skills,' 'Emotion-Related Skills,' 'Dealing with Aggressive Behavior and Urges Skills,' 'Problem-Solving Skills,' 'Planning, Group Interaction,' and 'Executing a Task Skills.' Each module's scores are summed and divided by the number of items in that module to obtain a score. A high score obtained from the scale, which does not have inversely coded items, indicates high social skills. Cronbach Alpha internal consistency coefficient is 0.97 for the test (Ataş, Efeçinar, and Tatar 2016).

Statistical analysis

Data were analyzed according to a normal distribution with SPSS 26.00 software. Continuous variables were summarised using mean and standard deviation, and categorical variables were summarised as frequency and percentage. Visual methods (histograms and probability graphs) and analytical methods (Kolmogorov–Smirnov/Shapiro–Wilk tests) analyzed the normality of distribution for continuous variables. An independent sample t-test was used to measure the difference between the groups, and linear regression analysis was used to examine the influential factors on the Social Skills Assessment Scale. Statistical significance value $p < .005$ was accepted.

Results

The socio-demographic factors of the children with ASD are presented in Table 1. The mean age of children with ASD who participated in physical activity was 12.33 ± 1.58 years, and the mean age of children with ASD who did not participate was 12.57 ± 1.52 years. The mean duration of physical activity participation of children with ASD who participated in physical activity was 2.03 ± 0.85 . Of the children with ASD who participated in physical activity, 20% were swimming, 16.7% running, 16.7% football, 16.7% table tennis, and 29.9% exercise activities with physical education teachers at the sports center.

The comparison results of the group's Autism Behavior Checklist and the Social Skills Assessment Scale for children averages are presented in Table 2. Accordingly, children in the group who participated in physical activity showed a statistically significant difference compared to the group who did not participate in physical activity ($p < .001$).

Demographic variables and the Autism Behavior Checklist predicting the Social Skills Assessment Scale were examined in the groups participating and not participating in physical activities. As a result of the linear regression analysis, it was determined that the predictive effect of demographic variables was not statistically significant. In addition, the predictive effect of the Autism Behavior Checklist on the Social Skills Assessment Scale in those who did not participate in physical activities was not statistically significant. In contrast, the predictive effect of the Autism Behavior Checklist on the Social Skills Assessment Scale in those who participated in physical activities was statistically significant (Table 3).

Table 1. Demographic characteristic of participants and their parents.

	Children with autism participating in physical activity			Children with autism not participating in physical activity		
	n	%	M $\pm$ SD	n	%	M $\pm$ SD
Age of child (year)			12.33 $\pm$ 1.58			12.57 $\pm$ 1.52
Gender						
Girl	16	53.3		10	33.3	
Boy	14	46.7		20	66.7	
Physical Activities						
Swimming	6	20				
Athletics	5	16.7				
Football	5	16.7				
Ping Pong	5	16.7				
Other	9	29.9				
How long have you been participating in physical activity? (year)			2.03 $\pm$ 0.85			–
Information about parents						
Age of parents (year)			40.13 $\pm$ 3.24			39.63 $\pm$ 3.38
Who answers the questionnaires?						
Mother	20	66.7		19	63.3	
Father	10	33.3		11	36.7	
Employments status of parents						
Employed	17	56.7		20	66.7	
Not employed	13	43.3		10	33.3	

Table 2. The results of independent samples t-test.

	Children with autism participating in physical activity		Children with autism not participating in physical activity		t	p
	M	SD	M	SD		
Autism Behavior Checklist - ABC	30.40	3.80	50.17	7.67	12.647	.001**
Social Skills Assessment Scale	2.40	0.27	1.67	0.16	12.759	.001**

** $p < .001$; * $p < .05$.

Table 3. Regression analysis of the predict of social skills on behavior skills.

		Unstandardized Coefficients		Standardized Coefficients Beta	t	p	95,0% CI	
		B	Std. Error				Lower Bound	Upper Bound
Children With Autism Not Participating in Sports	(Constant)	1.891	0.194		9.731	.001	1.493	2.289
	Autism Behavior Checklist	−0.004	0.004	−0.214	−1,159	.256	−0.012	0.003
(F:1,342, p:.256, R ² :0.046)								
Children With Autism Participating in Sports	(Constant)	3.967	0.285		13.925	.001	3.383	4.551
	Autism Behavior Checklist	−0.052	0.009	−0.723	−5.541	.001	−0.071	−0.032
(F:30,699, p:0 < .001, R ² :0.523)								
Dependent variable: Social Skills Assessment Scale, Independent variable: Autism Behavior Checklist								

Discussion

The most important result of this study, which was conducted to reveal the difference between social skills and behaviors in children with ASD who participate in active physical activity and those who do not, was the determination that children who do not participate in physical activity are more socially and behaviorally disadvantaged. The results of this study were significant for the families of children with ASD and individuals and professionals involved with these children. Encouraging children with ASD to participate in physical activity as much as possible can contribute to their social and behavioral development.

The initial finding of our study was that children with ASD who participate in active physical activity tend to have higher scores in social skills compared to those who do not participate in physical activity. Similar to Gregor et al. (2018) study with a comparable sample group, they anticipated that interventions for children diagnosed with ASD would predominantly focus on behavioral/communication interventions. They also suggested that studies focusing on physical activity would enhance social skills but noted families' reluctance. Yarımıkaya and Esentürk (2022) predicted that children with ASD experienced insufficient physical activity in the post-COVID-19 period. They foresaw that participation in physical activities could positively impact social and behavioral skills. Their study also provided examples of physical activities these children could participate in at home.

Jimeno (2019) indicated that athletic-based interventions conducted in a group setting result in gains in social functionality. Young and Furgai (2016) also stated that physical activity such as swimming, running, and trampoline jumping positively impacts the social skills of children with ASD. In the semi-experimental study conducted by Castillo and Dalmau (2021), children with ASD were taught the activity of riding a bicycle. It was reported that there was an increase in social skills, self-confidence, and motor skills at the end of the study. Cai et al. (2020) similarly stated that a 12-week basketball training program increased social skills. Our study, by the literature, consistently demonstrated, based on quantitative data, that children with ASD who participate in physical activity had higher social skills than those who do not. This may indicate the benefits that regular participation in physical activities can provide and that children with better social skills, those with mild autism levels, should be directed to physical activity. When completing the scales, parents reported that participating in physical activities increased their children's non-verbal communication skills, such as making eye contact, gestures, and facial expressions in the appropriate context. Therefore, it will be necessary for future studies to reveal the effectiveness of developmentally appropriate physical activity interventions on the social skills of children with ASD.

Another result of our study is that children with ASD who participate in active physical activity exhibit fewer negative behaviors such as aggression, hostility,

anger, and crying compared to those who do not participate in active physical activity. Young and Furgai (2016) noted the positive impact of physical activity on the behaviors of children with ASD. Gabriels et al. (2015) mentioned a significant improvement in behavioral skills related to nervousness and behavior in relationships in children with ASD through group-based horseback riding activities along with behavioral skills training. In our study, the enhancing effect of physical activity on behaviors in children with ASD was emphasized, which is consistent with the literature. On the other hand, Duffy et al. (2017) noted changes in emotional responses and repetitive behaviors through the running program they named 'Monday Club.' However, they could not attribute these changes to the program due to the absence of a control group. In our study, the behavioral skill scores of children with ASD who participated in active physical activity were compared to those who did not, and it was found that undesirable behaviors were less frequent in children who participated in physical activity. In addition to the results of the scales, parents reported that inappropriate behaviors such as shouting, self-harm, and stereotypical behaviors were becoming less common in their children. This situation may indicate both the benefits of physical activity and the redirection of children with more desirable behaviors toward physical activity activities.

Our study's last result was that the social skills prediction for behavioral skills differed between groups. Accordingly, the social skills of children with ASD who participated in physical activities predicted behavioral skills. However, the social skills of children with ASD who did not participate in physical activities did not predict behavioral skills. The difference between the two groups in terms of social and behavioral skills may mean that children with ASD who do not participate in physical activity are not yet able to use their social skills sufficiently or are not sufficiently developed (Lee, Chang, and Jolin 2021; Sansi, Nalbant, and Ozer 2021; Zhao and Chen 2018). This may be interpreted as their social skills not yet reflected in their behavior (Soto-Icaza et al. 2015). Moreover, it may be associated with the advantages of physical activity, such as increasing sociability and providing peer learning by entering a peer environment (Hynes and Block 2022). Future studies should contribute to the literature by examining the effect of physical activity in children with ASD in more detail.

This study has several strengths and limitations. One limitation is that the sample group was taken from a single center, which may limit the generalizability of

the study results. Additionally, children's reasons for participating in physical activities may be related to having better social skills and behavior patterns. Therefore, in future studies, it will be important to demonstrate the effectiveness of various physical activity interventions tailored to the development of children with ASD on social skills and undesirable behaviors. On the other hand, a strength of our study is its rarity in connecting these skills with physical activity in children with social and behavioral problems like ASD. Based on the data obtained from our study, it is recommended that future studies investigate how physical activity in various disciplines affects the social and behavioral skills of individuals with ASD. Future studies showing the different effects of regular physical activity in children with ASD will contribute to the literature. In addition, we recommend that mixed-methods studies, including qualitative research, should be conducted by taking the views of children who participate in regular physical activity and their parents to help us understand what has changed positively or negatively with the start of physical activity, especially social issues such as the impact of participation in physical activities on communication with peers and family members.

Conclusion

Overall, we can conclude that physical activity is associated with social skills and behaviors in children with ASD. Children with ASD who participate in physical activities have higher behavioral and social skills than those who do not. This study is important for raising awareness among parents of children with ASD, teachers, exceptional education instructors, and healthcare professionals by highlighting the importance of physical activity. In this context, we believe that policymakers and managers working in relevant departments should make necessary regulations to increase societal awareness and support access to physical activities for individuals with disabilities.

Authors contributions

Conceptualization: Gokcen Akyurek, Serkan Toplu; Methodology: Gokcen Akyurek, Serkan Toplu; Formal analysis and investigation: Gokcen Akyurek, Ezginur Gundogmus; Writing - original draft preparation: Ezginur Gundogmus; Writing - review and editing: Ezginur Gundogmus, Gokcen Akyurek; Supervision: Gokcen Akyurek

Disclosure statement

No potential conflict of interest was reported by the authors.

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Data availability statement

The data are available on request from the corresponding author.

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